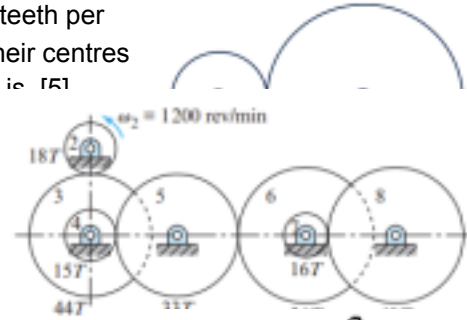


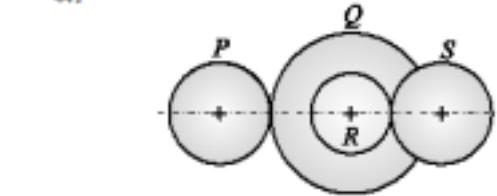
ME 150

Basics of Mechanical and Civil Engineering Assignment #3 (Max marks 70) (Gears and Belt drive)

1. Two meshing spur gears 1 and 2 with diametral pitch of 8 teeth per mm and an angular velocity ratio $|\omega_2|/|\omega_1| = 1/4$, have their centres 30 mm apart. The number of teeth on the driver (gear 1) is 16.



2. What are the speed and direction of rotation of gear 8? [5]



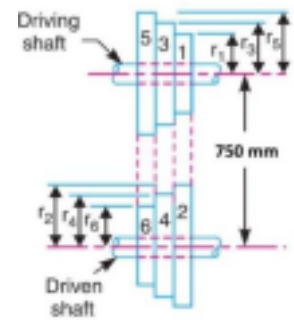
3. A compound gear train with gears P, Q, R and S has number of teeth 20, 40, 15 and 20 respectively. Gears Q and R mounted on the same shaft as shown in figure. The diameter of the gear Q is twice that of the gear R. If the module of the gear R is 2 mm, the centre distance in mm between gears P and S is [10]

4. A shaft which rotates at a constant speed of 160 r.p.m. is connected by belting to a parallel shaft 750 mm apart, which has to run at 60, 80 and 100

r.p.m. The smallest pulley on the driving shaft is 40 mm in radius. Determine

the remaining radii of the two stepped pulleys for an open belt, Neglect belt

thickness and slip. [15]



5. Determine the width of a 9.75 mm thick leather belt required to transmit 15 kW from a motor running at 900 r.p.m. The diameter of the driving pulley of the motor is 300 mm. The driven pulley runs at 300 r.p.m. and the distance between the center of two pulleys is 3 m. The density of the leather is 1000 kg/m^3 . The maximum allowable stress in the leather is 2.5 MPa. The coefficient of friction between the leather and pulley is 0.3. Assume open belt drive and neglect the sag and slip of the belt. [15]

6. The following data refer to an open belt drive: Diameter of larger pulley = 400 mm; Diameter of smaller pulley = 250 mm; Distance between two pulleys = 2m; Coefficient of friction between smaller pulley surface and belt = 0.4; Maximum tension when the belt is on the point of slipping = 1200 N. Find the power transmitted at speed of 10 m/s. It is desired to increase the power. Which of the following two methods you will select?

1. Increasing the initial tension in the belt by 10 per cent.
2. Increasing the coefficient of friction between the smaller pulley surface and belt by 10 per cent

by the application of suitable dressing on the belt. Find, also, the percentage increase in power possible in each case. [15]

7. Determine the maximum power that can be transmitted by a belt of 100 mm×10 mm size with an angle of lap of 160° . The belt density is 1000 kg/m³ and coefficient of friction is 0.25. The tension in the belt should not exceed 1.5 MPa. [5]